

Seminars 1 and 2: Sinusoids and Complex Exponentials

Topics: sine and cosine functions, sinusoidal amplitude, frequency, period, phase, time shift, sampling, complex numbers, Euler's formula, complex exponentials, rotating phasors, inverse Euler formulas, and phasor addition.

Key idea. A sinusoid is completely determined by its amplitude, frequency, and initial phase. Complex exponentials and phasors provide an efficient way to represent and combine sinusoids of the same frequency.

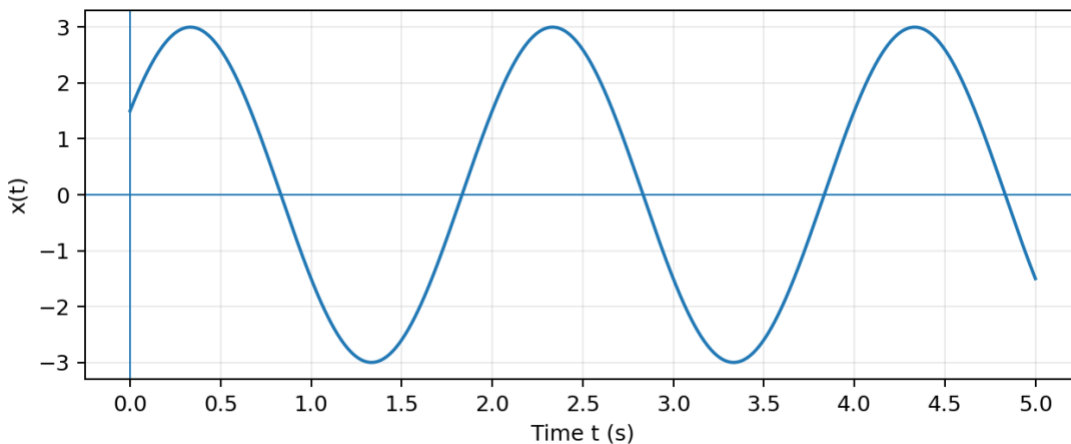
1. Review of sine and cosine

Make a carefully labelled sketch of each function. Mark the amplitude, period, maxima, minima, and zero crossings.

- $x_1(\theta) = \cos\theta$, for $0 \leq \theta \leq 6\pi$.
- $x_2(t) = 2\cos(0.4\pi t)$, showing three complete periods.
- $x_3(t) = 2\cos(0.4\pi t + \pi/2)$, showing three complete periods.
- $x_4(t) = 2\sin(0.4\pi t)$, showing three complete periods.
- Explain the relation between the graphs in parts (c) and (d).

2. Reading sinusoidal parameters from a graph

The following graph represents a sinusoid of the form $x(t) = A\cos(\omega_0 t + \phi)$.



Determine:

- the amplitude A ;
- the period T_0 ;
- the cyclic frequency f_0 in hertz;
- the radian frequency ω_0 in radians per second;
- one valid value of the initial phase ϕ , using the convention $-\pi < \phi \leq \pi$.

3. Frequency and period

For each signal, determine A , ω_0 , f_0 , T_0 , and ϕ . Express phase in the interval $-\pi < \phi \leq \pi$.

- $x_1(t) = 5\cos(6\pi t - \pi/4)$.
- $x_2(t) = 3\sin(20\pi t + \pi/6)$, first rewriting it as a cosine.

- (c) $x_3(t) = -4\cos(8t + 5\pi/3)$, using a positive amplitude.
- (d) $x_4(t) = 2\cos(120\pi t + 7\pi/2)$.

4. Phase and time shift

A sinusoid may be written in either form $x(t) = A\cos(\omega_0 t + \phi) = A\cos(\omega_0(t - t_1))$.

- (a) Derive the relation between ϕ and t_1 .
- (b) For $x(t) = 4\cos(10\pi t - 3\pi/4)$, determine the smallest positive delay t_1 .
- (c) Rewrite $x(t) = 3\sin(12\pi t)$ in delayed-cosine form.
- (d) Explain why the same sinusoid has infinitely many equivalent phase values and infinitely many equivalent time shifts.

5. Maxima, minima, and zero crossings

Consider $x(t) = 6\cos(8\pi t + \pi/3)$.

- (a) Find the first maximum at or after $t = 0$.
- (b) Find the first minimum at or after $t = 0$.
- (c) Find the first two positive zero crossings.
- (d) Verify your answers by sketching two periods of the signal.

6. Sampling a continuous-time sinusoid

Let $x(t) = 20\cos(80\pi t - 0.4\pi)$.

- (a) Write the sampled sequence $x[n] = x(nT_s)$ for $T_s = 0.005$ s.
- (b) How many samples are taken per period?
- (c) Calculate $x[n]$ for $n = 0, 1, 2, 3, 4$.
- (d) Repeat part (b) for $T_s = 0.0025$ s and explain which sampled plot would appear smoother.
- (e) Plot the continuous-time sinusoid and both sampled sequences over two periods.

7. Complex-number review for sinusoidal analysis

For each complex number, convert between Cartesian and polar/exponential form, and plot it in the complex plane.

- (a) $z_1 = 3 + j4$.
- (b) $z_2 = -2 + j2\sqrt{3}$.
- (c) $z_3 = 5e^{-j\pi/6}$.
- (d) $z_4 = 2e^{j5\pi/4}$.

Then compute $z_1 z_3$ and z_1/z_3 most conveniently in polar form.

8. Euler's formula and inverse Euler formulas

Starting from $e^{j\theta} = \cos\theta + j\sin\theta$,

- (a) derive $e^{-j\theta}$;
- (b) derive the inverse Euler formula for $\cos\theta$;
- (c) derive the inverse Euler formula for $\sin\theta$;
- (d) use the formulas to express $3\cos(5t - \pi/4)$ as the sum of two complex exponentials.

9. Rotating complex exponentials

Consider $z(t) = 2e^{j(\pi t + \pi/4)}$.

- Determine its magnitude and phase at $t = 0$.
- Determine the direction of rotation in the complex plane.
- Determine its period.
- Calculate and plot $z(t)$ at $t = 0, 0.25, 0.5, 0.75, 1$ s.
- Sketch $\Re\{z(t)\}$ and $\Im\{z(t)\}$ over one period.

10. Representing a real sinusoid with a phasor

For each signal, identify the phasor $X = Ae^{j\phi}$ and write a complex signal $z(t) = Xe^{j\omega_0 t}$ such that $x(t) = \Re\{z(t)\}$.

- $x_1(t) = 4\cos(100\pi t - \pi/3)$.
- $x_2(t) = 3\sin(40\pi t + \pi/4)$.
- $x_3(t) = -2\cos(60\pi t - \pi/6)$, using a positive phasor magnitude.

Plot the three phasors in the complex plane.

11. Addition of two same-frequency sinusoids

Use phasors to combine $x(t) = 2\cos(200\pi t + \pi/3) + 2\cos(200\pi t - 3\pi/4)$ into the form $x(t) = A\cos(200\pi t + \phi)$.

- Draw the two original phasors.
- Add them in Cartesian form.
- Convert the sum to polar form.
- State the resulting amplitude and phase.
- Verify the result at $t = 0$.

12. Addition of three same-frequency sinusoids

Use phasors to combine $x(t) = 2\cos(300\pi t) + 3\cos(300\pi t + \pi/2) + 4\cos(300\pi t - \pi)$ into one cosine.

- Draw the three phasors.
- Add them graphically and algebraically.
- Express the result as $A\cos(300\pi t + \phi)$.
- Explain what would change if one term had a different frequency.

13. General two-phasor addition identity

Let $X_1 = e^{j\alpha}$, $X_2 = e^{j\beta}$, $X_3 = X_1 + X_2$.

- Show that $X_3 = 2\cos\left(\frac{\alpha-\beta}{2}\right)e^{j(\alpha+\beta)/2}$.
- Interpret the magnitude and angle of X_3 geometrically.
- What happens when $\alpha - \beta = \pi$?
- What happens when $\alpha = \beta$?

14. Discrete-time complex exponential

Consider $x[n] = 2e^{j(\pi n/4 + \pi/6)}$, $n = 0, 1, \dots, 15$.

- State the magnitude, normalized radian frequency, and initial phase.

- (b) Determine the fundamental period in samples.
- (c) Plot $\Re\{x[n]\}$ and $\Im\{x[n]\}$ versus n .
- (d) Plot the samples $x[n]$ as points in the complex plane.
- (e) Explain why the points repeat.

15. Integrated computational exercise

Using Python, analyse $x(t) = 5\cos(2\pi 110t + \pi/3) + 4\cos(2\pi 110t - \pi/4)$.

- (a) Plot both component sinusoids and their sum over three periods.
- (b) Draw both phasors and their vector sum in the complex plane.
- (c) Compute the equivalent amplitude and phase numerically.
- (d) Plot the equivalent single sinusoid and verify that it coincides with the sum.
- (e) Sample the signal using at least 20 samples per period and display the samples together with the continuous curve.